

Wisest

Curriculum Database

A-level Further Mathematics — Complete Registry

Core Pure 1 & 2 · Further Statistics 1 & 2 · Further Mechanics 1 & 2

*Every subtopic is a record with a stable primary-key id,
name and coverage description.*

July 5, 2026

Supersedes Parts IV–V of the combined registry of July 3, 2026.
The GCSE and A-level Mathematics registries (Parts I–III) are unchanged.

Seed registry for the Wisest question-bank pipeline. Each id is the key against
which a 55-question bank is generated and verified.

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Schema

Each row below is one subtopic record. Fields:

id — primary key. Stable dot-slug, globally unique, never reused. This is the identifier every generation agent, ledger entry and bank file references.

subtopic — human-readable name.

description — one-line summary of the mathematics covered; the generation brief for that record.

Key grammar. *fm.year.strand.subtopic* with *year* ∈ *y1*, *y2* and *strand* ∈ *pure* (Core Pure), *stats* (Further Statistics), *mech* (Further Mechanics). The applied content follows the Further Statistics / Further Mechanics option route; the decomposition itself is board-independent across AQA, Edexcel and OCR. There is no tier field at this level.

Id stability

All 43 ids from the July 3 registry are retained with identical keys. Where a record has been split, the original id keeps the core of its former scope with a *narrowed* description, and the carved-out content lives under new sibling ids. New records are marked **NEW** in the tables. No id has been renamed, merged away or reused.

Registry summary

Year	Module	Records	(was)
Further Maths Y1 (33)	Core Pure 1	17	9
	Further Statistics 1	9	6
	Further Mechanics 1	7	4
Further Maths Y2 (37)	Core Pure 2	19	10
	Further Statistics 2	9	7
	Further Mechanics 2	9	7
Total		70	43

70 subtopic records × 55 questions each = 3,850 questions at full generation (previously 2,365). Combined with the unchanged GCSE (123) and A-level Mathematics (83) registries, the full database is 276 records and 15,180 questions.

Change log against the July 3 registry

Retained unchanged (25): all of FS1 except *fm.y1.stats.chi-squared*; all four original FM1 records except narrowings noted below; *fm.y1.pure.induction*, *fm.y1.pure.roots-polynomials*, *fm.y1.pure.series*, *fm.y1.pure.volumes-revolution*; *fm.y2.pure.roots-of-unity*, *fm.y2.pure.method-of-differences*, *fm.y2.pure.maclaurin*, *fm.y2.pure.first-order-des*, *fm.y2.pure.second-order-des*; all of FS2 except *fm.y2.stats.estimation-ci*; the FM2 circular-motion, centres-of-mass, statics and further-dynamics records.

Retained with narrowed description (18): split records — e.g. *fm.y1.pure.matrices* now covers arithmetic only, with determinants/inverses, linear systems and invariance carved out; *fm.y2.pure.de-moivre* now covers the theorem and powers, with exponential form, multiple-angle work and $C + iS$ series carved out; *fm.y2.mech.collisions-2d* now covers sphere-surface impact only, with sphere-sphere oblique impact carved out.

New (27): marked **NEW** throughout. Counts per module in the summary table above.

1 Further Maths Year 1: Pure (Core Pure 1)

Key prefix `fm.y1.pure.` — 17 records.

id	subtopic	description
<code>fm.y1.pure.complex-arithmetic</code>	Complex numbers: arithmetic	Four operations on complex numbers; complex conjugates; equating real and imaginary parts.
<code>fm.y1.pure.complex-quadratics</code> NEW	Complex roots of polynomials	Quadratics, cubics and quartics with real coefficients and complex roots; conjugate-pair arguments.
<code>fm.y1.pure.argand</code>	Argand diagrams	Points and vectors in the complex plane; modulus and argument; modulus-argument form.
<code>fm.y1.pure.argand-loci</code> NEW	Loci and regions	Loci $ z - a = r$, $ z - a = z - b $, $\arg(z - a) = \theta$; shading regions; extremising $ z $ and $\arg z$ on a locus.
<code>fm.y1.pure.roots-polynomials</code>	Roots of polynomials	Root-coefficient relations for quadratics to quartics; sums of powers of roots.
<code>fm.y1.pure.transforming-roots</code> NEW	Transforming roots	Forming new polynomials whose roots are linear transformations of the roots of a given polynomial.
<code>fm.y1.pure.induction</code>	Proof by induction	Induction for series results, divisibility and matrix powers.
<code>fm.y1.pure.series</code>	Sums of series	Standard results for $\sum r$, $\sum r^2$, $\sum r^3$; sums of related series by linearity and splitting.
<code>fm.y1.pure.matrices</code>	Matrix arithmetic	Addition, scalar and matrix multiplication; zero and identity matrices; non-commutativity.
<code>fm.y1.pure.determinants-inverses</code> NEW	Determinants and inverses	2×2 and 3×3 determinants and inverses; singularity; $\det(AB)$ and $(AB)^{-1}$ properties.
<code>fm.y1.pure.matrix-systems</code> NEW	Systems of linear equations	Matrix solution of three simultaneous equations; consistency; geometric configurations of three planes.
<code>fm.y1.pure.linear-transformations</code>	Linear transformations	Matrices for rotations, reflections, enlargements and stretches in 2D; rotations about coordinate axes in 3D; successive transformations; determinant as area/volume scale factor.
<code>fm.y1.pure.invariance</code> NEW	Invariant points and lines	Invariant points; invariant lines and lines of invariant points of 2D matrix transformations.
<code>fm.y1.pure.vectors-lines-planes</code>	Equations of lines and planes	Vector and Cartesian equations of lines in 3D and of planes, including the scalar-product form.
<code>fm.y1.pure.scalar-product</code> NEW	Scalar product methods	Angles between two lines, two planes, and a line and a plane; perpendicularity and parallelism conditions.
<code>fm.y1.pure.vector-distances</code> NEW	Intersections and distances	Intersections of lines and of a line with a plane; perpendicular distances point-line, point-plane and between skew lines; reflections.
<code>fm.y1.pure.volumes-revolution</code>	Volumes of revolution	Rotation about the x - and y -axes; composite regions; modelling.

2 Further Maths Year 1: Statistics (Further Statistics 1)

Key prefix `fm.y1.stats.` — 9 records.

id	subtopic	description
<code>fm.y1.stats.discrete-rv</code>	Discrete random variables	Expectation and variance; $E(aX + b)$, $\text{Var}(aX + b)$; functions of X .

continued

id	subtopic	description
fm.y1.stats.poisson	The Poisson distribution	Conditions; mean and variance; additivity; approximation to the binomial.
fm.y1.stats. geometric-negbinomial	Geometric and negative binomial	Trials to first and r th success; means and variances.
fm.y1.stats.hypothesis-tests	Hypothesis testing (Poisson, geometric)	Tests for Poisson and geometric parameters; critical regions.
fm.y1.stats.central-limit	Central limit theorem	Distribution of sample means for large samples; approximate inference.
fm.y1.stats.chi-squared	Chi-squared goodness of fit	Goodness of fit to discrete uniform, binomial, Poisson and geometric models; pooling classes; degrees of freedom.
fm.y1.stats. contingency-tables NEW	Contingency tables	Chi-squared tests of independence; expected frequencies; degrees of freedom $(r - 1)(c - 1)$.
fm.y1.stats.pgfs NEW	Probability generating functions	PGFs of standard discrete distributions; mean and variance from $G(t)$; sums of independent variables.
fm.y1.stats. quality-of-tests NEW	Quality of tests	Type I and Type II errors; size and power of a test; power functions.

3 Further Maths Year 1: Mechanics (Further Mechanics 1)

Key prefix fm.y1.mech. — 7 records.

id	subtopic	description
fm.y1.mech.momentum-impulse	Momentum and impulse	Impulse–momentum principle and conservation of momentum for direct motion in a line.
fm.y1.mech.impulse-vectors NEW	Impulse and momentum in 2D	Impulse and momentum as vectors; impact and jerk problems in two dimensions.
fm.y1.mech.work-energy-power	Work, energy and power	Work, kinetic and potential energy; work–energy principle; power, including on inclines.
fm.y1.mech. elastic-strings-springs	Elastic strings and springs	Hooke’s law for strings and springs; tension and thrust; equilibrium with elastic components.
fm.y1.mech.elastic-energy NEW	Elastic energy	Elastic potential energy; energy conservation and the work–energy principle with elastic strings and springs.
fm.y1.mech.collisions-1d	Elastic collisions in one dimension	Newton’s law of restitution; direct impact of spheres and impact with a fixed surface; kinetic energy loss.
fm.y1.mech. successive-collisions NEW	Successive collisions	Successive direct impacts between spheres and with walls; conditions for further collisions.

4 Further Maths Year 2: Pure (Core Pure 2)

Key prefix fm.y2.pure. — 19 records.

id	subtopic	description
fm.y2.pure. complex-exponential NEW	Exponential form	$re^{i\theta}$ form; multiplication and division; the geometric effect of multiplying by a complex number.

continued

id	subtopic	description
fm.y2.pure.de-moivre	De Moivre's theorem	Statement and proof by induction; integer and rational powers of complex numbers.
fm.y2.pure.multiple-angles NEW	Multiple-angle identities	Expressing $\cos n\theta$, $\sin n\theta$ in powers of trig functions and vice versa; applications to equations and integration.
fm.y2.pure.trig-series NEW	Trigonometric series	Sums of trigonometric series via geometric series of complex exponentials ($C + iS$ method).
fm.y2.pure.roots-of-unity	Roots of unity	n th roots of unity and of any complex number; geometric interpretation and regular polygons.
fm.y2.pure. method-of-differences	Series: method of differences	Telescoping sums, including with partial fractions.
fm.y2.pure.maclaurin	Maclaurin series	Series for e^x , $\ln(1+x)$, $\sin x$, $\cos x$, $(1+x)^n$; validity; series of composites.
fm.y2.pure.further-calculus	Improper integrals and mean value	Improper integrals with infinite limits or unbounded integrands; the mean value of a function.
fm.y2.pure. inverse-trig-calculus NEW	Inverse trigonometric calculus	Differentiating inverse trig functions; integrals reducing to arcsin and arctan forms.
fm.y2.pure. further-integration NEW	Further integration	Integration of rational functions via partial fractions with quadratic factors; harder substitutions.
fm.y2.pure. volumes-parametric NEW	Volumes: parametric and modelling	Volumes of revolution for parametrically defined curves; adding and subtracting volumes; modelling.
fm.y2.pure.polar-coordinates	Polar coordinates	Polar–Cartesian conversion; sketching curves $r = f(\theta)$.
fm.y2.pure.polar-areas NEW	Polar areas and tangents	Areas enclosed by polar curves via $\frac{1}{2} \int r^2 d\theta$; tangents parallel and perpendicular to the initial line.
fm.y2.pure. hyperbolic-functions	Hyperbolic functions	Definitions, graphs and identities of $\sinh$, $\cosh$, $\tanh$; inverses and their logarithmic forms; equations.
fm.y2.pure. hyperbolic-calculus NEW	Hyperbolic calculus	Differentiating and integrating hyperbolic and inverse hyperbolic functions; integrals yielding arsinh , arcosh forms.
fm.y2.pure.first-order-des	First-order differential equations	Integrating factors; families of solution curves; modelling.
fm.y2.pure.second-order-des	Second-order differential equations	Constant-coefficient linear equations; complementary function and particular integral.
fm.y2.pure.de-modelling	Modelling with differential equations	Simple harmonic motion; damped and forced oscillations.
fm.y2.pure.coupled-des NEW	Coupled differential equations	Coupled first-order simultaneous linear equations; reduction to second order; interacting-population models.

5 Further Maths Year 2: Statistics (Further Statistics 2)

Key prefix fm.y2.stats. — 9 records.

id	subtopic	description
fm.y2.stats.linear-regression	Linear regression	Least-squares model, residuals and RSS; interpreting fitted models.

continued

id	subtopic	description
fm.y2.stats.continuous-rv	Continuous random variables	PDFs and CDFs; expectation, variance, median, mode; functions of X .
fm.y2.stats.continuous-uniform	The continuous uniform distribution	Properties, moments and applications of the rectangular distribution.
fm.y2.stats.combining-rv	Combinations of random variables	Means and variances of linear combinations of independent variables, including normal combinations.
fm.y2.stats.unbiased-estimators NEW	Unbiased estimators	Estimators and sampling distributions; bias; unbiased estimators of μ and σ^2 ; comparing estimators by variance.
fm.y2.stats.estimation-ci	Confidence intervals	Confidence intervals for a mean using the normal and t -distributions.
fm.y2.stats.variance-inference NEW	Inference for variance	Chi-squared confidence intervals and hypothesis tests for a population variance.
fm.y2.stats.further-tests	Further hypothesis tests	Tests for the difference of means; F -test for equality of variances; choosing an appropriate test.
fm.y2.stats.spearman	Spearman's rank correlation	Rank correlation coefficient and associated tests; when to prefer rank methods to the PMCC.

6 Further Maths Year 2: Mechanics (Further Mechanics 2)

Key prefix fm.y2.mech. — 9 records.

id	subtopic	description
fm.y2.mech.horizontal-circular	Horizontal circular motion	Angular speed; acceleration $r\omega^2$; conical pendulum and banked tracks.
fm.y2.mech.vertical-circular	Vertical circular motion	Energy methods on vertical circles; conditions for complete circles vs leaving the circle.
fm.y2.mech.com-laminas	Centres of mass: laminas	Uniform and composite laminas by symmetry and moments; remainder (cut-out) bodies.
fm.y2.mech.com-frameworks NEW	Centres of mass: frameworks	Frameworks of rods and wires, including circular arcs; composite frameworks by moments.
fm.y2.mech.com-calculus	Centres of mass: solids and calculus	Standard solids and solids of revolution by integration.
fm.y2.mech.statics-rigid	Statics of rigid bodies	Equilibrium of rigid bodies including suspension; toppling vs sliding on a plane.
fm.y2.mech.collisions-2d	Oblique impact with a surface	Oblique impact of a smooth sphere with a fixed surface; restitution normal to the surface; deflection angles and KE loss.
fm.y2.mech.oblique-spheres NEW	Oblique impact of spheres	Oblique impacts of two smooth spheres; restitution along the line of centres; successive oblique impacts.
fm.y2.mech.further-dynamics	Further dynamics	Motion under variable forces; $a = v \frac{dv}{dx}$ forms; inverse-square gravitation.